
A Capable, Fully Tensor-Decomposable Vision-Language-Action Policy

Exact rational conversion, protocol-aligned control, and weight-derived causal
structure

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Tensor decomposition factors a large table of interactions into a small set of low-
2 rank components that can be ranked and inspected. Applying it directly to a
3 conventional robot policy is difficult because softmax, ordinary activations, and
4 normalization interrupt the required algebraic form. We introduce χ -VLA, a com-
5 pact vision-language-action policy whose learned map uses only bilinear, linear,
6 and rational operations. Its checkpoint therefore defines an explicit rational tensor
7 program, making interactions among image, instruction, state, and action coordi-
8 nates accessible from the weights. A mechanical audit finds no hidden incompatible
9 operation, and local reconstructions agree near the float32 precision floor. On all
10 ten LIBERO-Object tasks, the 20.1M-parameter policy reaches $92.8\% \pm 2.1\%$
11 with a ViT encoder and $93.7\% \pm 0.8\%$ with a convolutional encoder under the
12 published three-seed protocol. We derive an exact weight-based decomposition
13 of softmax-free attention. All 36 primary block-head tests favor its rank-128 sub-
14 space over equal-rank random controls, with median fidelity ratio 2.44. At the
15 policy’s visual bond, retaining 64 of 384 causally ranked dimensions preserves
16 18/20 closed-loop successes, compared with 19/20 for the full representation and
17 0/20 for a random subspace. An 80-trial instruction intervention still exposes a
18 scene-identity shortcut. The evidence establishes competitive specialist capability
19 and exact layerwise structure, not broad generalization, negligible cost against a
20 matched conventional twin, or selective coefficient-level repair.

21 1 Introduction

22 Modern robot policies can work well while remaining difficult to inspect. A checkpoint contains
23 millions of learned numbers, but it does not directly reveal which combinations of image regions,
24 instruction tokens, and robot state produce an action. Analysts therefore usually collect activations
25 on a dataset and fit a separate probe. That can identify correlations, but it leaves open whether the
26 same structure is present in the weights and whether it causes behavior.

27 Tensor decomposition offers a different analysis surface. A tensor is a multidimensional table, and a
28 decomposition rewrites that table as a sum of simpler low-rank factors, much as a matrix factorization
29 finds dominant directions in a matrix. If a network is built from additions, multiplications, and ratios
30 of polynomials, its weights define an explicit table of input interactions. Decomposing that table can
31 rank the combinations the model is constructed to use. By *fully tensor-decomposable*, we mean that
32 every learned block preserves this algebraic form. The aim is not merely compression. It is a direct
33 path from parameters to candidate mechanisms that can be tested by intervention [3, 4].

Robot control makes this goal useful and difficult. Visual shortcuts can produce high benchmark success while ignoring the instruction, and small action errors compound in closed loop. Standard softmax, nonlinear activations, and square-root normalization break the explicit polynomial form. Continuous actions also cannot rely on the external categorical sampling used by a language model. A practical design must recover these functions without losing precise control.

We study the narrower but testable question:

Can a specialist VLA retain competitive closed-loop capability when every deployed learned operation is polynomial or rational, and does the resulting structure provide a useful weight-based analysis surface?

The stronger thesis has three parts: decomposability should have small capability cost, the weights should expose stable behaviorally causal interactions, and editing those interactions should selectively change behavior without retraining. This paper establishes strong capability and exact layerwise structure, then states clearly where causal diagnosis and editing remain incomplete.

Our contributions are:

1. We construct a complete VLA from tensor-compatible primitives. A rational normalization supplies input-specific scaling without leaving the rational function class.
2. We evaluate two fully rational 20.1M-parameter policies under a protocol aligned with published LIBERO-Object results. The strongest single architecture reaches $93.7\% \pm 0.8\%$ over three seeds, and a three-checkpoint ensemble reaches 96.2%.
3. We mechanically audit the real trained policy and explicitly reconstruct deployed rational and bilinear branches from their weights.
4. We extend exact weight-derived decomposition through individual attention layers with a reduced-Gram calculation that avoids materializing the full high-order tensor.
5. We show a causal closed-loop consequence. A selected 64-dimensional visual subspace preserves almost all tested behavior, while a random equal-width subspace fails. A separate instruction intervention exposes a real scene-identity shortcut and limits the language-grounding claim.

Table 1 separates the evidence already established from the tests still needed for the stronger thesis. This distinction is important because algebraic representability does not by itself imply tractable analysis, and a causally important activation subspace is not yet a coefficient-level editing interface.

Conjunct	Current evidence	Decisive missing test
Near-zero capability cost	Two fully rational 20.1M policies reach 92.8% and 93.7% mean success.	Same-skeleton conventional and partial-conversion twins under paired budgets.
Weight-only diagnosis	Exact layerwise attention decomposition consistently beats fixed random subspaces.	Recover a concrete behavioral condition from weights before viewing its test rollouts.
Predictive surgery	A selected activation subspace causally preserves behavior.	Edit specified coefficients, predict the direction, and measure target effect and collateral damage without retraining.

Table 1: **Status of the three-part thesis.** Strong capability and exact layerwise structure are established. The matched cost and coefficient-surgery claims remain open.

This paper does not claim current state of the art. It also does not claim that comparable mechanisms are impossible to obtain from conventional policies. The empirical question is whether the algebraic policy makes them more direct, stable, and selectively editable than matched post-hoc methods.

67 2 A rational tensor robot policy

68 2.1 Tensor-compatible primitives

69 Each primitive below replaces a familiar transformer component while keeping an explicit coefficient description. Linear maps contribute first-order terms. Multiplying learned linear projections
70 creates higher-order interactions whose coefficients remain functions of the weights, rather than an
71 unstructured nonlinear black box.
72

73 **Homogeneous coordinate.** We write $\bar{x} = [1; x]$. The constant coordinate allows a multiplicative
74 layer to represent bias, linear, and quadratic terms in one contraction.

75 **Bilinear feed-forward block.** For $L, R \in \mathbb{R}^{r \times (d+1)}$ and $D \in \mathbb{R}^{d \times r}$,

$$\text{BFFN}(x) = D [(L\bar{x}) \odot (R\bar{x})], \quad y_o = \sum_{i,j} \left(\sum_{\rho} D_{o\rho} L_{\rho i} R_{\rho j} \right) \bar{x}_i \bar{x}_j. \quad (1)$$

76 The inner sum is an order-three table indexed by output coordinate and two input coordinates. The
77 three matrices give its CP factorization, the tensor analogue of a low-rank matrix factorization,
78 without materializing every table entry.

79 **Softmax-free bilinear attention.** Each head forms two query-key systems and one value stream:

$$Y = C [M \odot (Q_1 K_1^\top) \odot (Q_2 K_2^\top) / s] V. \quad (2)$$

80 Here M is a fixed causal mask, C is fixed row scaling, and s is fixed. All learned query, key, value,
81 and output maps are linear. Equation 2 is therefore a polynomial in its input activations. Unlike
82 softmax attention, every multiplicative interaction can be expanded into coefficients determined by
83 the trained weights.

84 **Rational per-instance normalization.** RMS normalization contains an input-dependent square
85 root. We instead deploy a fixed rational approximation $r(z) = P(z)/Q(z)$ to $1/\sqrt{z + \epsilon}$:

$$\text{RNorm}(x) = x r \left(\frac{1}{d} \sum_{j=1}^d x_j^2 \right). \quad (3)$$

86 Here *rational* means a ratio of two polynomials. Projective coordinates carry each activation as a
87 numerator-denominator pair (N, D) representing N/D . Linear, bilinear, residual, and normalization
88 operations update this pair. The model therefore keeps input-specific rescaling while remaining
89 exactly representable as a rational tensor network.

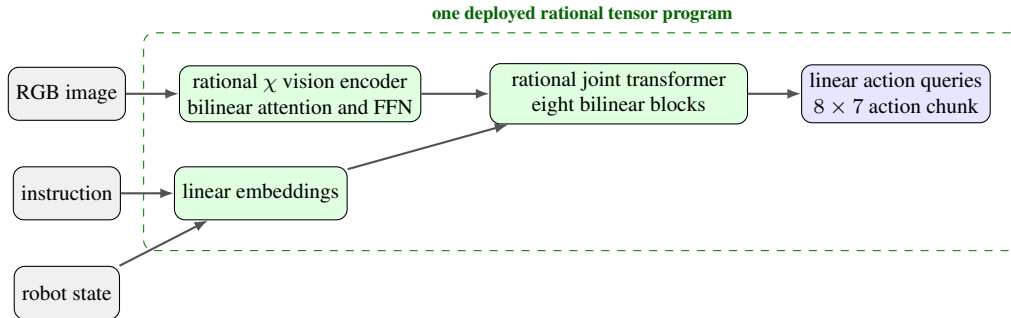


Figure 1: **Deployed χ -VLA map.** The strongest checkpoints use one vision stream and a linear continuous-action head. Every learned component inside the dashed boundary is linear, bilinear, or rational.

2.2 Architecture and training

The policy follows the high-level VLA sequence of visual encoding, language and state embedding, joint processing, and action decoding [1]. It uses a 64×64 agent-view image, an eight-dimensional robot state, a 26-token task vocabulary, and eight action-query tokens. The joint backbone has width 384, eight blocks, and twelve heads. The action head maps each action-query state directly to seven continuous action dimensions. The complete model has 20,137,352 learned parameters.

Training uses 66,984 demonstration frames, 40,000 AdamW updates, batch size 256, cosine decay, bfloat16 arithmetic, and an exponential moving average. Closed-loop evaluation uses a 180° camera rotation to match the demonstration convention, canonical LIBERO initial states, and fail-loud checks that verify every requested state was loaded.

2.3 Mechanical certificate on the trained checkpoint

Architecture definitions can differ from deployed code. We therefore audit the real 189/200 checkpoint at runtime and reconstruct representative operations from saved weights.

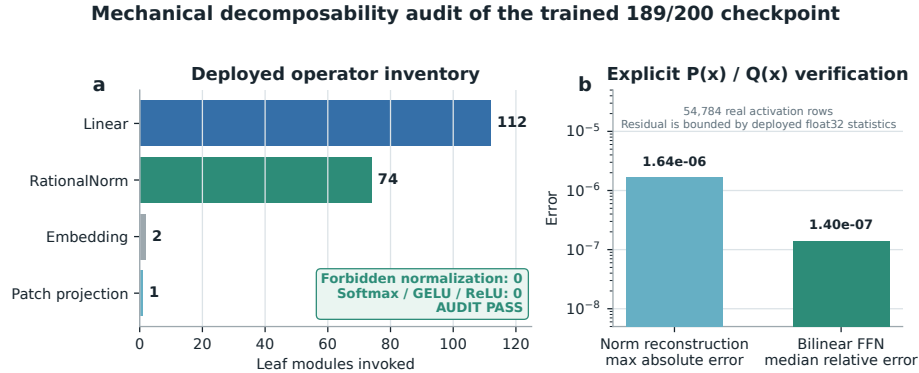


Figure 2: **Mechanical audit of the trained rational checkpoint.** The runtime path invokes 112 linear maps, 74 rational normalizers, two embeddings, and one linear patch projection. No softmax, standard activation, or ordinary normalization is present. Reconstruction errors are measured over 54,784 real activation rows.

The normalization reconstruction has maximum absolute error 1.64×10^{-6} . A bilinear FFN reconstruction has median relative error 1.40×10^{-7} . These residuals are bounded by the deployed module’s float32 statistic. The audit certifies operator membership and local reconstruction. It does not materialize one compact polynomial for the full eight-layer transformer.

3 Protocol-aligned closed-loop capability

3.1 Evaluation protocol

We evaluate all ten LIBERO-Object tasks with:

- 50 canonical trials per task
- a 280-step cap
- three independently trained seeds per architecture
- 500 rollouts per seed and 1,500 rollouts per architecture

Published OpenVLA and Diffusion Policy results are protocol-aligned references. They were not rerun in our codebase, so differences may still include implementation and training choices.

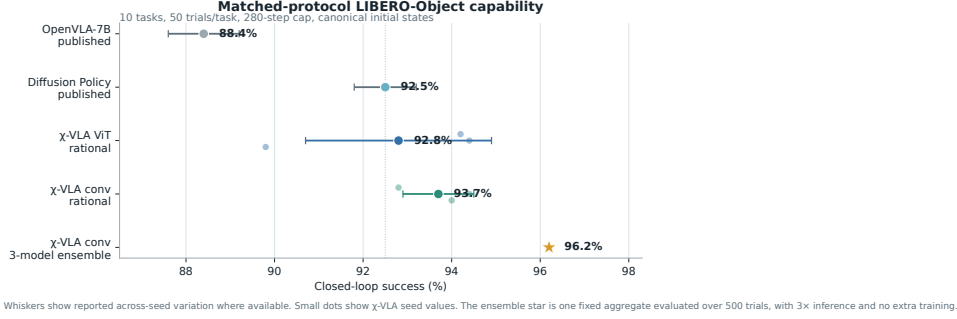


Figure 3: **Protocol-aligned LIBERO-Object capability.** Small dots are χ -VLA training seeds. The ensemble star averages three convolutional checkpoints and uses three forward passes per action chunk. External references are reported published values.

3.2 Results

Method	Parameters	Training seeds	Success
χ -VLA, rational ViT	20.1M	3	$92.8\% \pm 2.1\%$
χ -VLA, rational conv	20.1M	3	$93.7\% \pm 0.8\%$
χ -VLA, conv ensemble	$3 \times 20.1M$	$3 \rightarrow 1$	96.2%
OpenVLA-7B [1]	7B	3	$88.4\% \pm 0.8\%$
Diffusion Policy [1]	not matched	3	$92.5\% \pm 0.7\%$

Table 2: **Closed-loop success under aligned evaluation settings.** The two external rows are historical references rather than current state-of-the-art claims.

The ViT seeds score 89.8%, 94.4%, and 94.2%. The convolutional seeds score 94.0%, 94.4%, and 92.8%. The convolutional model is modestly higher and more consistent in this three-seed comparison.

Elementwise averaging of the three convolutional action predictions reaches 96.2%. This is 2.3 points above the mean of its members and 1.2 points above the strongest member. The gain is largest on tomato sauce, butter, and ketchup. The ensemble requires three complete forward passes and is not one 20.1M-parameter policy.

3.3 What this result does and does not establish

The result shows that the rational architecture supports high closed-loop specialist capability. It does not isolate the cost of decomposability. A decisive cost experiment must train a same-skeleton conventional transformer and staged partial-conversion controls using identical data, updates, seeds, and evaluation states. We treat that experiment as required before restoring the stronger title claim that convertibility is nearly free.

4 Exact weight-derived structure through attention

4.1 Exact layerwise construction

Equation 2 contains two query-key products and one value stream. Expanding the linear maps gives a high-order coefficient table. Each entry says how strongly one combination of query, key, and value coordinates contributes to an output. It is *weight-derived* because no examples or fitted probe are needed to construct it. On a tiny four-token, width-four, one-head layer, direct evaluation and an independent explicit polynomial agree to 2.00×10^{-15} . The internal core identity agrees to 1.24×10^{-14} .

The deployed attention also normalizes each query and key with Equation 3. Each normalization multiplies all channels of one token by the same scalar rational factor. Those factors can be pulled

outside the bilinear dot products. The polynomial attention core is unchanged, while explicit numerator and denominator factors carry the normalization. The rational extension agrees to 3.61×10^{-16} . Agreement with the real trained module ranges from 3.10×10^{-7} to 1.03×10^{-5} because the deployed implementation intentionally casts the statistic to float32.

Naively materializing the real width-384 coefficient tensor would require approximately 3.2×10^{15} values. We instead contract the CP factors into an $O(d^2)$ reduced Gram. This smaller matrix plays the role of a covariance matrix: its leading eigenvectors rank the directions that carry the most coefficient energy without constructing the full table. At toy multi-head scale, it matches brute-force materialization to 2.13×10^{-14} .

4.2 Trained policy result

We apply the exact weight-derived calculation to all twelve heads in early, middle-late, and final backbone blocks 0, 6, and 7. Random projector banks are sampled once, then fixed and reused for all evaluation examples. The weights choose the subspace. Real cached activations are used only afterward to measure how faithfully that subspace preserves the layer’s computation.

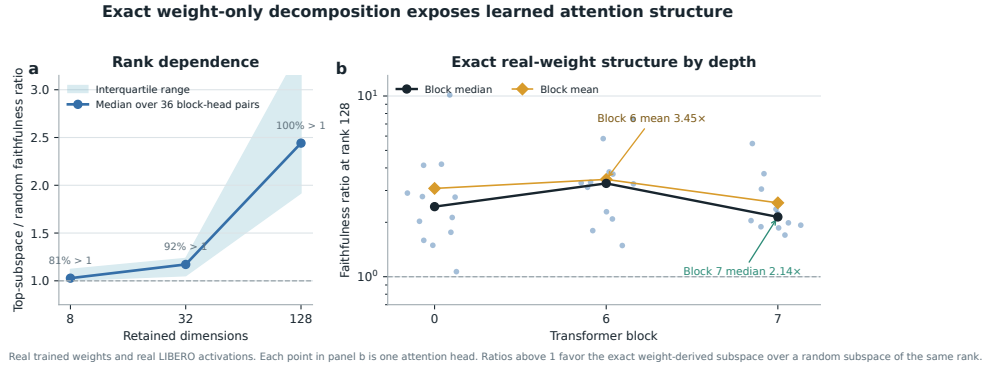


Figure 4: **Exact weight-derived attention structure.** Ratios above one favor the exact weight-derived subspace over a random equal-rank subspace. Each point in the right panel is one trained attention head.

At ranks 8, 32, and 128, respectively, 80.6%, 91.7%, and 100% of the 36 block-head pairs favor the exact subspace. Their median ratios are 1.028, 1.171, and 2.441. At rank 128, the mean ratio is 3.033 and the strongest head reaches 10.135. The rank-128 block means are 3.08, 3.45, and 2.56 for blocks 0, 6, and 7.

This extends exact weight-only construction through individual attention layers. It is not an exact end-to-end decomposition of the complete policy. Composition across all eight blocks still faces rapid degree and representation growth.

5 Causal policy structure and a language shortcut

We next test whether the identified structure affects closed-loop behavior and whether high task success reflects robust use of the language input.

5.1 Causal visual subspace

At the visual bond before the joint backbone, we rank directions separately for translation, rotation, and gripper actions. A subspace is a smaller coordinate system inside the model’s 384-dimensional visual representation. Retaining a subspace means zeroing every direction outside it, then running the actual policy. We compare the selected directions with random equal-width controls in offline reconstruction and closed-loop intervention. This ranking uses held-out activations and downstream sensitivity, so it tests a causal bottleneck but is not a weight-only discovery result.

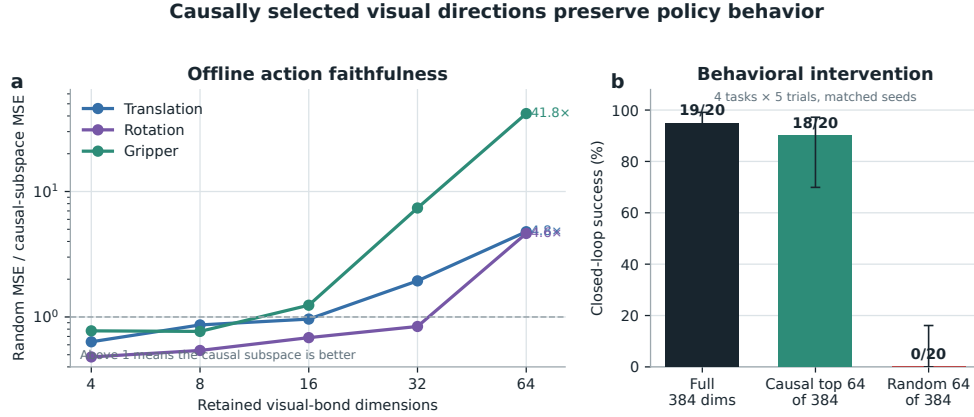


Figure 5: **Causally selected visual directions preserve policy behavior.** The left panel reports random MSE divided by selected-subspace MSE. The right panel intervenes on the real policy over four tasks and five paired trials per task.

171 With 64 of 384 dimensions retained, the selected subspace reduces action reconstruction error relative
 172 to random by 4.8 $\times$ for translation, 4.6 $\times$ for rotation, and 41.8 $\times$ for the gripper. In closed loop, the
 173 full representation succeeds in 19/20 trials, the selected subspace in 18/20, and a random subspace
 174 in 0/20.

175 This result establishes a causal low-dimensional substrate, but it is still limited to one checkpoint and
 176 twenty rollouts. The current ranking at this full-policy bond uses held-out examples and downstream
 177 sensitivity. The exact layerwise attention method in Section 4 must still be connected to the same
 178 full-suite closed-loop intervention.

179 5.2 Counterfactual instruction test

180 High success does not guarantee robust language use. We hold the image and robot state fixed
 181 and replace the instruction with one naming another visible object. Across 80 canonical one-step
 182 trials, the predicted reach direction shifts toward the newly named object in only 15.0%. Mean
 183 directional cosine to the named object falls from 0.525 under the true instruction to 0.299 under the
 184 counterfactual instruction, while cosine to the original target remains 0.464. In an earlier 40-step
 185 closed-loop screen, 33.3% finish closer to the new object and 55.6% move measurably toward it.

186 The likely shortcut is structural in the dataset. Each scene template is paired with an almost fixed target,
 187 so scene identity can substitute for object-name grounding. Decorrelated scripted demonstrations
 188 improve the counterfactual metric, but all three fine-tuning attempts severely reduce ordinary closed-
 189 loop success. The best repaired dataset raises capability only to 21.5%, far below the healthy
 190 range.

191 This negative result narrows the VLA claim. The current model is a capable language-conditioned
 192 specialist, but it is not yet a robust instruction-grounded policy. LIBERO-CF independently documents
 193 the same broad failure class in existing VLAs [10]. Our prospective novelty is therefore not the
 194 shortcut alone. It is the harder test of whether exact weight-derived terms can localize and selectively
 195 repair that shortcut.

196 6 Decisive remaining tests

197 **1. Controlled conversion cost.** Train a same-skeleton conventional model with softmax attention,
 198 a standard gated FFN, and RMSNorm. Add partial-conversion controls. Use paired seeds and
 199 predeclare a three-point noninferiority margin. Match observations, token layout, action decoder,
 200 demonstrations, augmentation, optimizer, updates, evaluation states, and hyperparameter-search
 201 budget. Report parameters, FLOPs, throughput, memory, and latency. This is the missing experiment
 202 for any claim that convertibility is nearly free.

2. Direct coefficient surgery. Our first preregistered four-task screen directly edits query, key, and value columns selected by exact attention Grams. Retention gives a favorable 10-point gap over random controls but misses the 15-point gate, while removal has the wrong sign. This is a no-go for the current edit mapping, as detailed in Appendix A.5. The next test must resolve redundant paths or gauge sensitivity, compare post-hoc controls across three checkpoints, and report targeted effect, ordinary success, edit norm, and collateral damage.

3. Concrete diagnosis and selective repair. Develop two capstones. Gripper timing is the lower-risk target because phase transitions and premature closure have explicit simulator variables. Scene identity versus instruction binding is the higher-upside target because the current policy demonstrably fails the counterfactual test. In both cases, discover the algebraic term from weights, attach a limited semantic label, freeze the candidate, predict the behavioral consequence, and test on blind paired rollouts. A compelling repair should improve the targeted failure by 10 to 20 points, lose no more than 2 to 3 points of ordinary success, and preserve its direction across three seeds.

4. Broader capability and grounding. Evaluate the conventional and rational finalists on LIBERO-Goal and LIBERO-Long, followed by a counterfactual grounding benchmark such as LIBERO-CF [10]. Object-only success is insufficient because standard LIBERO can reward scene-to-action memorization [9]. Test the stability of recovered mechanisms under equivalent reparameterizations and across trained seeds.

5. RL only after a paired gate. The repaired-reward AWR pilot improves 80.7% to 82.7%, but it has one seed and fifteen trials per task. A justified follow-up needs three paired seeds, a matched-compute behavior-cloning continuation control, full protocol evaluation, and regression gates. Generic PPO on saturated LIBERO-Object is lower priority than the four tests above.

7 Related work

VLA capability. OpenVLA established an open pretrained VLA and standardized LIBERO comparisons [1]. OpenVLA-OFT showed that continuous action chunks and simple regression can substantially improve adaptation and control efficiency [2]. These systems use large pretrained backbones. χ -VLA instead studies a compact from-scratch specialist under an algebraic restriction.

Tensor and polynomial networks. Polynomial networks and tensor networks use explicit multilinear structure for representation and learning [12–14]. Bilinear MLP work showed that weight eigendecomposition can expose low-rank features and circuits [3]. χ -nets introduced ODT for compositional bilinear networks [4]. We extend the setting to a continuous closed-loop policy and give an exact reduced construction for individual softmax-free attention layers.

Causal mechanisms and policy analysis. Bilinear IMPALA policies have combined competitive ProcGen control with weight-derived eigenfilters, followed by activation probing and control [5]. This is the closest architectural precedent for weight-based policy analysis. Recent work steers pretrained VLAs through causally identified activation directions [6]. Sparse feature circuits identify and edit causal feature graphs in conventional language models [7], while nonlinear causal reductions recover behavioral patterns in reinforcement-learning policies from rollout perturbations [8]. These results make a universal separation from conventional policies untenable. Our intended distinction is empirical: exact weight-derived discovery and direct coefficient edits in a competitive continuous vision-language policy, evaluated for stability and collateral damage against matched post-hoc baselines.

VLA robustness. Recent evaluations show that standard LIBERO success can coexist with weak robustness and language use [9–11]. Our instruction intervention reaches the same conclusion from an independently trained compact specialist and motivates a decomposition-guided diagnosis.

8 Limitations and conclusion

χ -VLA demonstrates that a complete rational tensor policy can reach strong specialist closed-loop capability. The real checkpoint passes a mechanical operator audit. Exact weight-derived analysis

now crosses individual attention layers, and a compact causal visual subspace preserves almost all tested behavior.

Five limits determine the next version of the paper. First, no same-skeleton conventional twin has yet measured the exact capability and compute cost of conversion. Second, the benchmark is one narrow suite and the policy fails a counterfactual instruction test. Third, exact decomposition is layerwise rather than end to end. Fourth, the closed-loop causal intervention uses one checkpoint and twenty trials. Fifth, activation projection succeeds but the first direct coefficient-edit screen fails its selectivity gate, so no behaviorally validated weight edit has yet been established.

The defensible conclusion is therefore narrow. Polynomial and rational restrictions do not prevent a compact policy from reaching competitive LIBERO-Object capability, and they provide a precise weight-derived analysis surface. Establishing negligible conversion cost, broad language-grounded control, semantic diagnosis from weights, and a decomposition-enabled repair requires the controlled experiments in Section 6. If all three conjuncts in Table 1 hold on the same trained policies, the result is stronger than any individual capability, decomposition, or intervention claim.

Reproducibility. Training and evaluation use fixed task lists, explicit camera and action conventions, canonical-state manifests, checkpointed moving-average weights, and per-run result files. The final artifact will release code, checkpoint hashes, exact attention certificates, and per-episode evaluation logs. Preflight checks fail if canonical initial states cannot be loaded.

References

- [1] M. J. Kim et al. *OpenVLA: An Open-Source Vision-Language-Action Model*. CoRL, 2024.
- [2] M. J. Kim, C. Finn, and P. Liang. *Fine-Tuning Vision-Language-Action Models: Optimizing Speed and Success*. 2025.
- [3] A. Rigg, T. Doods, M. T. Pearce, J. Oramas, and L. Sharkey. *Bilinear MLPs Enable Weight-Based Mechanistic Interpretability*. ICLR, 2025.
- [4] T. Doods, W. Gauderis, G. A. Wiggins, and J. Oramas. *Compositionality Unlocks Deep Interpretable Models*. 2025.
- [5] F. Oozeer, Y. Erisken, A. Rigg, J. Oramas, and L. Sharkey. *Bilinear Convolution Decomposition for Causal RL Interpretability*. 2024.
- [6] B. Häon, K. Stocking, I. Chuang, and C. Tomlin. *Mechanistic Interpretability for Steering Vision-Language-Action Models*. 2025.
- [7] S. Marks et al. *Sparse Feature Circuits: Discovering and Editing Interpretable Causal Graphs in Language Models*. ICLR, 2025.
- [8] M. Kekić et al. *Learning Nonlinear Causal Reductions to Explain Reinforcement Learning Policies*. ICLR, 2026.
- [9] X. Zhou et al. *LIBERO-PRO: Towards Robust and Fair Evaluation of Vision-Language-Action Models Beyond Memorization*. 2025.
- [10] Y. Fang et al. *When Vision Overrides Language: Evaluating and Mitigating Counterfactual Failures in VLAs*. 2026.
- [11] C. Kim et al. *LIBERO-Para: A Diagnostic Benchmark and Metrics for Paraphrase Robustness in VLA Models*. 2026.
- [12] T. Kolda and B. Bader. *Tensor Decompositions and Applications*. SIAM Review, 2009.
- [13] E. Stoudenmire and D. Schwab. *Supervised Learning with Tensor Networks*. NeurIPS, 2016.
- [14] G. Chrysos et al. *Deep Polynomial Neural Networks*. CVPR and TPAMI, 2020 to 2021.

294 A Supporting results and decisions

295 A.1 Original 94.5% capability result

296 Before the final protocol-aligned benchmark, one rational ViT checkpoint succeeded in 189/200 trials
 297 under a 600-step horizon and 20 trials per task. This result established that the rational implementation
 298 could support all ten tasks. It should not be compared directly with the protocol-aligned external
 299 references because it uses one training seed, fewer trials, and a longer horizon.

300 A.2 Per-task ensemble behavior

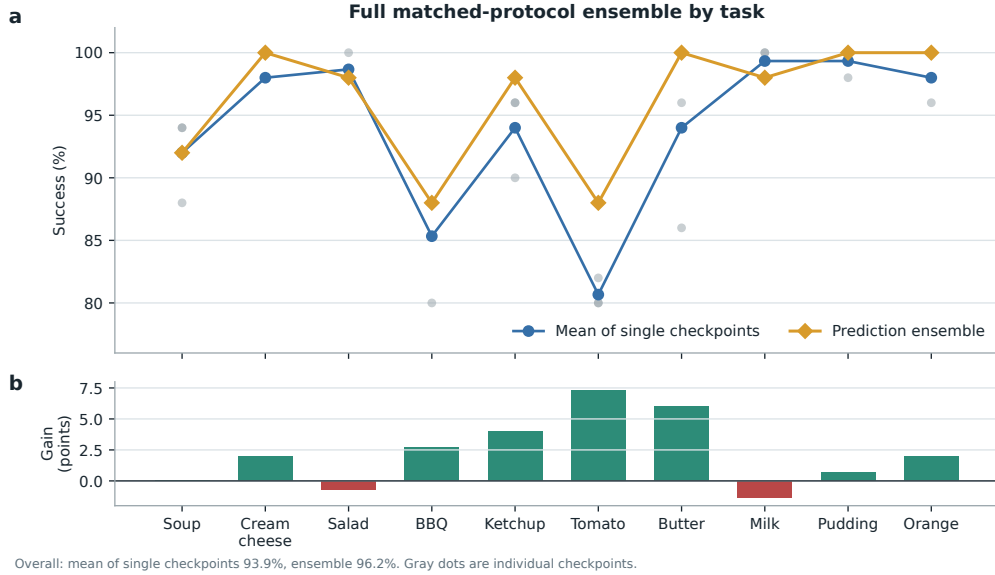


Figure 6: **Per-task effect of prediction averaging.** Gray points are the three convolutional checkpoints, evaluated with 50 canonical trials per task. Curves compare their taskwise mean with elementwise action-prediction averaging. The ensemble raises overall success from a 93.9% member mean to 96.2%, with its largest gains on tomato sauce, butter, and ketchup. It requires three forward passes and is not a single 20.1M-parameter policy.

301 A.3 Product-routing multimodal head

302 For an action-query state h , the product-routing head represents

$$a(h, b) = c_0(h) + \sum_{g=1}^G (2b_g - 1)c_g(h), \quad b \in \{0, 1\}^G. \quad (4)$$

303 The center, offsets, and gates are bilinear CP modules. A final external sign choice selects one of
 304 2^G modes. The head folds to numerical precision, but its five-seed closed-loop mean is $44\% \pm 30\%$
 305 and its range overlaps the linear baseline. The strongest rational policy therefore uses the linear head.
 306 Product routing remains a construction result until it wins a controlled genuinely multimodal task.

307 A.4 Development interventions

308 Focused DAGger on BBQ and tomato improves combined success from 82.5% to 96.0% using seven
 309 corrective episodes and 1,131 frames. Applying the same update thinly across all ten tasks reduces
 310 overall success from 92.0% to 72.0%. Three original AWR runs decline by 8.0, 4.6, and 2.6 points.
 311 A reward audit shows that accumulated step cost makes every successful trajectory have negative
 312 return. After removing that term and repairing the terminal boundary, one pilot improves by two
 313 points. These are development diagnostics, not validated paper contributions.

What the July 30 intervention experiments taught us

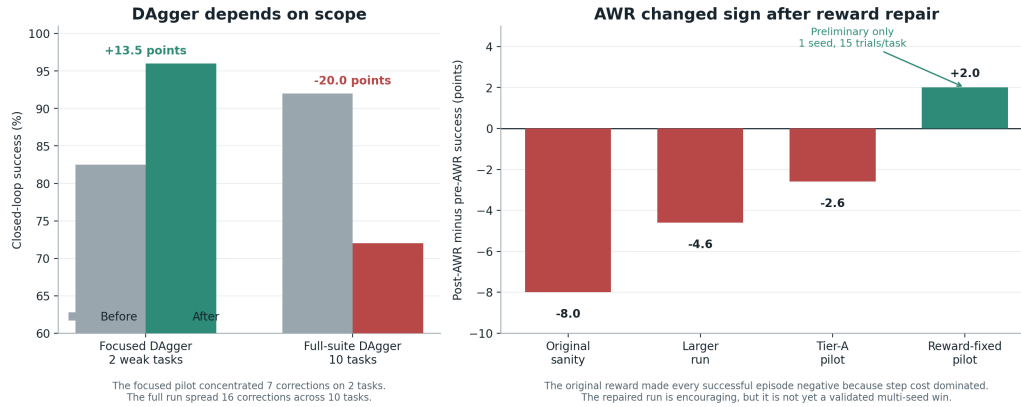


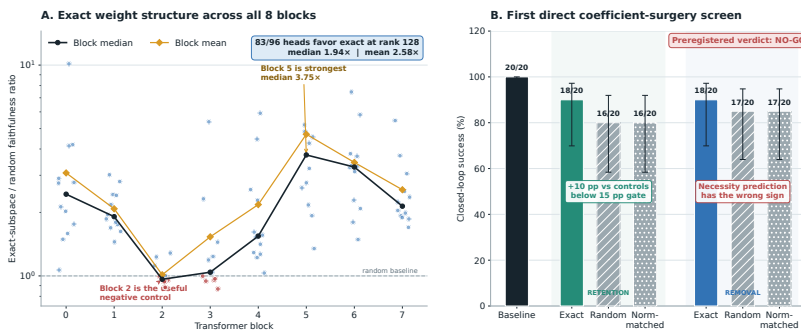
Figure 7: **Development interventions reveal scope and reward sensitivity.** Focused DAGger helps two weak tasks, while spreading a small correction budget across ten tasks causes regression. Three AWR variants also regress before a reward audit and repair. The positive reward-repaired AWR result is a one-seed pilot with 15 trials per task, so it is not evidence of a validated improvement.

314 A.5 Expanded attention screen and first direct edit

315 The corrected fixed-control attention analysis covers all eight blocks and twelve heads. At rank 128,
 316 83/96 block-head pairs favor the exact weight-derived subspace. The median ratio is 1.94, the mean
 317 is 2.58, and blocks 2 and 3 provide useful weak or negative controls. At ranks 8 and 32, only 54/96
 318 and 60/96 pairs favor the exact subspace. The effect is therefore depth-dependent and strongest at
 319 the wider tested rank.

320 The first direct weight-edit screen uses four tasks and five paired canonical episodes per task. Exact-
 321 subspace retention preserves 18/20 successes compared with 16/20 for both random controls. This
 322 10-point advantage misses the preregistered 15-point gate. Exact removal also preserves 18/20,
 323 compared with 17/20 for both random controls, which is the wrong direction for the necessity
 324 prediction. The screen is a no-go for the current mapping from Gram eigenspaces to query, key, and
 325 value column edits.

Weights expose reproducible structure, but the first direct edit is not yet selective



A: 96 block-head pairs, fixed random-control banks, rank 128. Exact directions come from weights; cached activations only score layer faithfulness. B: 4 tasks × 5 paired canonical episodes. Direct Q/K/V weight edits, no hooks, gradients, retraining, or rollout fitting. Error bars are 95% Wilson intervals.

Figure 8: **Expanded weight-structure and direct-surgery results.** Left: all 96 block-head pairs at rank 128, with fixed random-control banks. Exact directions come from weights, while cached activations only score layer faithfulness. Right: the first direct query-key-value column-edit screen over four tasks and five paired trials per task. Error bars are 95% Wilson intervals. The preregistered verdict is no-go because retention misses the effect-size gate and removal has the wrong sign.

326 **A.6 Numerical scope of rational conversion**

327 The deployed Padé computation is exactly rational even though it approximates RMS normalization.
328 Final certification must report its polynomial degrees, coefficients, denominator margin, observed
329 activation range, tail error, action drift, and runtime overhead. Operator membership alone does not
330 establish practical numerical safety under distribution shift.